Cayden Hutcheson

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Education

UT San Antonio May 2028

B.S. Computer Science, Minor in Mathematics, Specialization in Data Science *GPA: 4.0/4.0*

- **Honors:** 2x President's List, 4x Experiential Learning Award, PURF Fellowship
- **Organizations:** Honors College, ACM, ICPC, Cyber Jedi's

Experience

Major League Hacking September 2026 – Present

Coach *United States*

- Mentor 2,000+ student developers monthly at international hackathons on full-stack, AI/ML, and cloud builds.
- Partner with event organizers on technical infrastructure, developer blockers, and end-to-end event logistics.
- Build sponsor demos (GitHub Copilot, Google Gemini, Roo Code) to drive adoption and judge advanced usage.
- Design and host 6+ hours/month of workshops on data structures and ML, reaching 5,000+ developers.

MITRE June 2026 – Present

Systems Engineer Intern *San Antonio, TX*

- Cut workflow time 34% on USAF systems-modeling platforms by building Okapi BM25 log-parsing workflows.
- Removed manual intervention from a deployed system by automating that pipeline end to end in Python.
- Built a Python/FastAPI and React pipeline that turns requirements docs and Cameo models into SysML v2.
- Converted a 2,994-element Cameo model to 822 lines of SysML v2 with 0 parser errors and no Cameo license.
- Leading the technical report comparing AI-generated and hand-built systems models for the MBSE team.

Pumps of Houston May 2026 – Present

Software Engineer Intern *Houston, TX*

- Independently built a bid-triage platform that grades ∼45 public bids/month by parsing spec PDFs with an LLM.
- Turned vendor quote PDFs into Acumatica ERP sales orders with LLM extraction and per-customer pricing.
- Cut ERP invoice sync from 16.5 minutes to 80 seconds by batching writes across 13,500+ invoices.
- Shipped a Next.js staff dashboard and customer portal over 36 Postgres tables, backed by 1,200+ tests.

UT San Antonio October 2025 – Present

AI/ML Research Assistant, iTEA Lab *San Antonio, TX*

- Mapped 10TB+ of Reddit data to the Gartner Hype Cycle by building NLP pipelines over 1.7B+ text samples.
- Enabled near-instant retrieval across that corpus by automating analysis with topic modeling and semantic search.
- Collected 3,354 posts and comments from 9 subreddits; BERTopic surfaced 28 GenAI-critique themes.
- Leading an LLM Socratic physics tutor for middle schoolers with a RAG misconception matrix; targeting CHI.

UT San Antonio August 2025 – Present

Resident Assistant, Chisholm Hall *San Antonio, TX*

- Supported 150+ residents through community events, policy enforcement, and on-call incident response.
- Cover night and weekend on-call shifts, responding to building emergencies and resident crises.
- Mediate roommate conflicts and refer residents to campus counseling, academic, and health resources.
- Plan and advertise community programs, and document policy, safety, and maintenance issues in reports.
- Coded the captions, sound cues, title cards, and credits for the 12-person RA staff's 3-minute video.

Better Futures Institute January 2026 – May 2026

Software Engineer Intern *San Antonio, TX*

- Prioritized city infrastructure spending for a nonprofit by independently building pothole-detection models.
- Aligned 100+ stakeholders on one technical roadmap by directing AI implementation across a 15-person team.
- Built Python pipelines cleaning census, CDC, 311, and unemployment data into 2,021 RAG-ready rows.
- Added MongoDB response caching, query analytics, and Groq call logging to the civic chatbot's FastAPI backend.
- Wrote CDC PLACES, Census, and city open-data API loaders and grounded chatbot answers with source links.

MD Anderson Cancer Center December 2025 – May 2026

AI/ML Research Intern *Houston, TX*

- Raised bolus-type classification accuracy from 63% to 96% (95% macro F1) on 2,752 swallow-study videos.
- Trained and tuned I3D video models in PyTorch on A100 GPUs, finding 64-frame sampling performed best.
- Built the preprocessing pipeline that segmented, labeled, and split 277 swallow studies from 206 patients.
- Saved ∼20 researcher hours per week by automating feature extraction and model training; deployed clinically.

Technical Skills

Languages: Python, Java, TypeScript/JavaScript, Swift, SQL, HTML/CSS
Web & Cloud: React, Next.js, Three.js, FastAPI, REST APIs, AWS (EC2)
Data & Tooling: PostgreSQL, Docker, Kubernetes, GitHub Actions, Git/GitHub, Jira, Linux, SysML v2
AI/ML & Data: PyTorch, scikit-learn, NLP, RAG, embeddings, computer vision, LLM document extraction, Power BI, MCP
Clearance: Top Secret

Projects

Radar | GitHub | *Python, Swift, FastAPI, YOLO11, MongoDB* VTHacks 2026

- Won 2nd in GoDaddy's Agent Name Service track and 1st in UI/UX with an agent-gated home camera system.
- Built the entire ANS trust layer: 7 agents that verify each claim against the producer's key before acting.
- Wrote the SwiftUI iOS and watchOS apps, a wrist-to-hub incident flow over WatchConnectivity and Bonjour.
- Gated the camera's servo shutter on signed attestations, so the lens opens only on a verified grant.

Larp City | GitHub | *TypeScript, PixiJS, Node.js, PostgreSQL* HackRice 2026

- Won the Capital One Finance Track with a financial-literacy city sim where players live from 22 to retirement.
- Wrote its traffic simulation, save-and-rewind system, and Blender-to-pixel-art pipeline on a 3-person team.
- Rendered 6 hand-made cities across every US state in PixiJS, with routed cars, buses, and pedestrians.
- Added Gemini coaching, a news digest, and ElevenLabs narration; deployed on Express/Postgres.

Vision | Website | GitHub | *React, Three.js, Python, FastAPI* March 2026 – Present

- Selected as 1 of 15 US teams from 24,000 applicants (0.06%) for Red Bull Basement on a shipped prototype.
- Put structure, comparables, and zoning in one view by building React/Three.js 3D over FastAPI endpoints.
- Wrote a Python structural engine: beam moment, shear, and L/360 deflection over 6 ASCE 7-22 combos.
- Added roof, garage, and schedule build-along views to the 3D home viewer, over 65 FastAPI routes.
- Led the Red Bull Basement finalist build, adding CAD import, builder requests, and client-side caching.

Cyber Net | GitHub | *React, SQL, JavaScript, LLM APIs* RowdyHacks 2025

- Won 1st place, Best Cyber Hack, by building an autonomous penetration-testing platform with LLM detection.
- Ranked vulnerabilities by exploit risk by combining CVSS, EPSS, CISA KEV, and MITRE ATT&CK data.
- Turned scanned hosts into multi-step attack playbooks with runnable commands, beating 95+ teams in 24 hours.

Leadership

RowdyHacks | Website December 2025 – Present
Co-Director *San Antonio, TX*

- Raised $45,000+ from 10+ sponsors for a 900+ attendee hackathon by co-leading 5 organizing teams.
- Lead PR and logistics for a 24-hour hackathon, running sponsor calls with 5+ companies and MLH.
- Built sponsor decks and a data-driven impact report used to pitch renewals to returning sponsors.

Project Sunshine | Website October 2025 – Present
Lead Web Developer *San Antonio, TX*

- Improved volunteer onboarding for a pediatric-patient nonprofit by leading full-stack development of its site.
- Built the chapter's 7-page React/TypeScript site as sole developer, with a synced events calendar.
- Shipped a volunteer points system (7 activity types) and data files officers can edit without code.

Association for Computing Machinery (ACM UTSA) September 2025 – December 2025
Social Officer *San Antonio, TX*

- Organized and led technical workshops on programming and cyber awareness for 1,000+ student members.
- Planned 5+ large-scale events per semester with fellow officers, including hackathons and guest speakers.
- Created marketing content and ran ACM's social media to boost event engagement and membership.

Publications

Mao S., Sahli A.J., Buoy S.N., **Hutcheson C.**, et al. "Automated identification of bolus types in modified barium swallow studies using deep learning: a preliminary study." *medRxiv*, 2026 [Preprint]. doi:10.64898/2026.05.16.26353385